

Math Exam-01 — Attempt Overview

Units 1, 2 · Real Numbers, Logarithms · Seemab · 2026-04-18

SINGLE ATTEMPT

65 / 65

100% · PERFECT SCORE · Grade A+

12 / 12

Section A — MCQs
100%

33 / 33

Section B — Short
100%

20 / 20

Section C — Long
100%

What Made This Perfect

- **MCQs 12/12 clean** — every definition, property, and log-value recalled correctly including natural log (base e), characteristic of log 456.7, and scientific notation of 0.000052.
- **All 5 properties of real numbers** stated with correct general forms (Closure, Commutative, Associative, Additive Identity, Additive Inverse).
- **Radical operations** — both $\sqrt[3]{6 \cdot \sqrt{5}} = \sqrt[3]{30}$ and $\sqrt{20} \div \sqrt{5} = 2$ simplified cleanly.
- **Radical equation verified** — Q.2(v) solved $\sqrt{(3x+1)}=5$ to $x=8$, *and* substituted back to check. Textbook discipline.
- **Logarithm laws applied across 4 distinct questions** — find x, characteristic, combine into single log, digit-count of 2^{20} . No rote mistakes on any.
- **Algebraic identity recognized** — Q.3(i) elegantly applied $(a-b)(a+b) = a^2 - b^2$ to make the product-of-exponents collapse to 0.
- **Two digit-count problems both correct** — 2^{20} (7 digits) and 5^{37} (26 digits), using the "char + 1" rule consistently.
- **Bar-notation for negative characteristic** — Q.2(ix)(b) correctly wrote 3 for -3 , the Pakistani convention.

Where Nothing Went Wrong

No deductions on this paper. No incomplete sub-parts, no mislabelled elements, no arithmetic slips, no rushed endings. This is a textbook-clean attempt and a dramatic improvement on the pattern of "rushed long-answer sub-parts" flagged in the Chemistry exam-02 report (2026-04-17). Lesson learned and applied in a single day.

Key Techniques Seemab Demonstrated

Technique	Where Shown
Exponent rule $b^{m/n} = \sqrt[n]{(b^m)}$	Q.2(iv) OR
Squaring-both-sides for radical equations + verification	Q.2(v)
Characteristic + mantissa structure of log	Q.2(ix) OR, Q.2(xi), Q.3(iv) OR
Inverting $\log_x$ equation to solve for base	Q.2(vii) OR (b)

Factoring a composite number for logs: $72 = 2^3 \cdot 3^2$	Q.3(iii)
Difference-of-squares in exponent	Q.3(i) OR
Digit-count formula (digits = char + 1)	Q.2(xi), Q.3(iv) OR

Position in Track Record

This is Seemab's **6th exam** so far and her **first perfect score**.

#	Subject	Chapters	Date	Score	%
1	Biology	Ch 1-6	2026-04-03	52 / 65	80.0%
2	Physics	Ch 1-2	2026-04-07	63.5 / 65	97.7%
3	Physics	Ch 3-4, 9	2026-04-10	62 / 65	95.4%
4	Chemistry	Ch 1-2	2026-04-12	64 / 65	98.5%
5	Chemistry	Ch 3-5	2026-04-17	59.5 / 65	91.5%
6	Math	Units 1-2	2026-04-18	65 / 65	100%

New running average: 93.8% (up from 92.6% pre-attempt). Math becomes Seemab's strongest-scoring subject on first attempt. MCQ record now **71/72 = 98.6%**.

Report date: 2026-04-18 · **Attempt source:** math/exams/exam-01-ch1-2/attempts/2026-04-18/scans.pdf · **Detailed marking:** same dir/report.pdf